

PBPK evaluation report — Dapagliflozin

⚠ POOR FIT — overall GMFE 2.468 (>2-fold typical error). The model is not yet an adequate description of the data.

Objective: Build a whole-body PBPK model for Dapagliflozin that reproduces the observed plasma concentration-time data. Decide the model structure and any parameters you cannot obtain from the given data, and justify them. No modeling strategy and no reference results are provided. Overall GMFE 2.468 (reference 1.413); within-2-fold 44.1%. Structure: Cellular partition coefficient method - PK-Sim Standard, Cellular permeability - PK-Sim Standard; processes UGT1A9, UGT2B7, GlomerularFiltration, Hepatic-CYP.

Model, parameters & rationale

MODEL CHOICE

Distribution was described with Cellular partition coefficient method - PK-Sim Standard, Cellular permeability - PK-Sim Standard. The active processes were UGT1A9, UGT2B7, GlomerularFiltration, Hepatic-CYP. Parameters that cannot be reliably transferred from in-vitro to in-vivo were estimated against the clinical data; measured physicochemical inputs were fixed.

PARAMETERS

CLspec/[Enzyme]@UGT1A9 = 0.3946448242915066 [estimated] sens=1 ref=0.399443557
CLspec/[Enzyme]@UGT2B7 = 0.018850874596835872 [estimated] sens=0.081
collinear~Permeability(0.89) ref=0.0066043366201
Permeability = 2.8904303298160844e-05 cm/min [estimated] sens=0.13
collinear~CLspec/[Enzyme]@UGT2B7(0.89) ref=0.00037527645658
Lipophilicity = 2.0 [fixed] ref=2.6719093089
Fraction unbound (plasma, reference value) = 0.09 [fixed] ref=None
Solubility at reference pH = 100.0 [fixed] ref=0.2210041453
Reference pH = 7.0 [fixed] ref=None
Molecular weight = 408.873 [fixed] ref=None
GFR fraction = 1.0 [fixed] ref=0.7899801465
CLspec/[Enzyme]@Hepatic-CYP = 0.005 [fixed] ref=0.1432967727
Specific intestinal permeability (transcellular) = 0.001 cm/min [fixed] ref=3.9684694792e-05
Cl = 1.0 [held-at-default] ref=None
^ not fitted and carries no literature origin - an unvalidated default the agent could have fitted but did not

PHARMACOLOGICAL RATIONALE

Each estimated parameter, checked against independent literature where available:

- CLspec/[Enzyme]@UGT1A9 = 0.3946 Sensitivity 1: well constrained by the data.
- CLspec/[Enzyme]@UGT2B7 = 0.01885 Sensitivity 0.081 (of 1.0): the fit barely responds to this parameter – weakly identifiable, so its absolute value is uncertain.
- Permeability = 2.89e-05 cm/min – within the physiological range [1e-06, 1.0]. Sensitivity 0.13: moderately constrained by the data.

Model equations (ODE system)

COMPARTMENTS / STATE VARIABLES

PK-Sim's standard human whole-body structure: ~15 perfused organs (lung, brain, heart, kidney, gut, spleen, pancreas, liver, stomach, muscle, skin, bone, adipose, gonads) plus arterial and venous blood pools. Each organ is further divided into plasma, blood-cell, interstitial and intracellular sub-compartments, so the solved system is one mass-balance ODE per sub-compartment (order $\sim 10^2$ states). The physiological volumes V and flows Q are fixed by the individual; only the drug terms below are model choices.

GOVERNING EQUATIONS

Perfused tissue (distribution):

$$V_t \cdot dC_t/dt = Q_t \cdot (C_{art} - C_t / (K_{p,t} / R_{bp}))$$

(Each non-eliminating organ t (brain, heart, muscle, skin, adipose, bone, ...): rate in from arterial blood at flow Q_t , rate out with the venous effluent in equilibrium with tissue via the tissue:plasma partition coefficient $K_{p,t}$ (R_{bp} = blood:plasma ratio).)

Venous blood pool:

$$V_{ven} \cdot dC_{ven}/dt = \sum_t Q_t \cdot C_t / (K_{p,t} / R_{bp}) - Q_{co} \cdot C_{ven} + R_{iv}(t)$$

(Collects the effluent of every organ; Q_{co} = cardiac output. IV doses enter here as an input rate $R_{iv}(t)$ (bolus or infusion).)

Arterial blood pool:

$$V_{art} \cdot dC_{art}/dt = Q_{co} \cdot (C_{lung} - C_{art})$$

(Blood is oxygenated/mixed through the lung and distributed to the organs.)

Permeability-limited exchange (cellular):

$$V_{cell} \cdot dC_{cell}/dt = P \cdot SA \cdot (f_u \cdot C_{int} - f_{u,cell} \cdot C_{cell} / K_{p,t})$$

(Where cellular permeability limits uptake, the intracellular space is a separate state exchanging with interstitium across the membrane (permeability $P \times$ surface area SA), driven by the unbound gradient.)

Liver (hepatic metabolism):

$$V_{liv} \cdot dC_{liv}/dt = Q_{liv} \cdot C_{art} + Q_{po} \cdot C_{po} - Q_{hv} \cdot C_{liv} / (K_{p,liv} / R_{bp}) - CL_{int} \cdot f_u \cdot C_{liv}$$

(The liver receives arterial + portal (Q_{po} , first-pass) inflow; CYP3A4 metabolism removes drug as a first-order intrinsic-clearance sink $CL_{int} \cdot f_u \cdot C_{liv}$ on the unbound concentration.)

Kidney (renal filtration):

$$V_{kid} \cdot dC_{kid}/dt = Q_{kid} \cdot (C_{art} - C_{kid} / (K_{p,kid} / R_{bp})) - GFR \cdot f_u \cdot C_{kid}$$

(Glomerular filtration removes unbound drug at rate $GFR \cdot f_u \cdot C_{kid}$.)

Oral absorption (PO doses):

$$dA_{lum}/dt = -P_{int} \cdot SA_{int} \cdot C_{lum,u}; \quad J_{abs} = P_{int} \cdot SA_{int} \cdot C_{lum,u} \rightarrow \text{gut wall} \rightarrow \text{portal vein} \rightarrow \text{liver}$$

(Dissolved drug in the intestinal lumen permeates the mucosa (specific intestinal permeability $P_{int} \times$ surface area), enters the gut wall, and drains via the portal vein to the liver (first-pass).)

Model equations (ODE system)

TISSUE PARTITIONING

$K_{p,t}$ is not a fitted number but is computed per organ from tissue composition by the PK-Sim Standard method – a function of effective lipophilicity, the unbound fraction f_u , drug ionization (pKa) and each tissue's lipid / water / phospholipid / protein content.

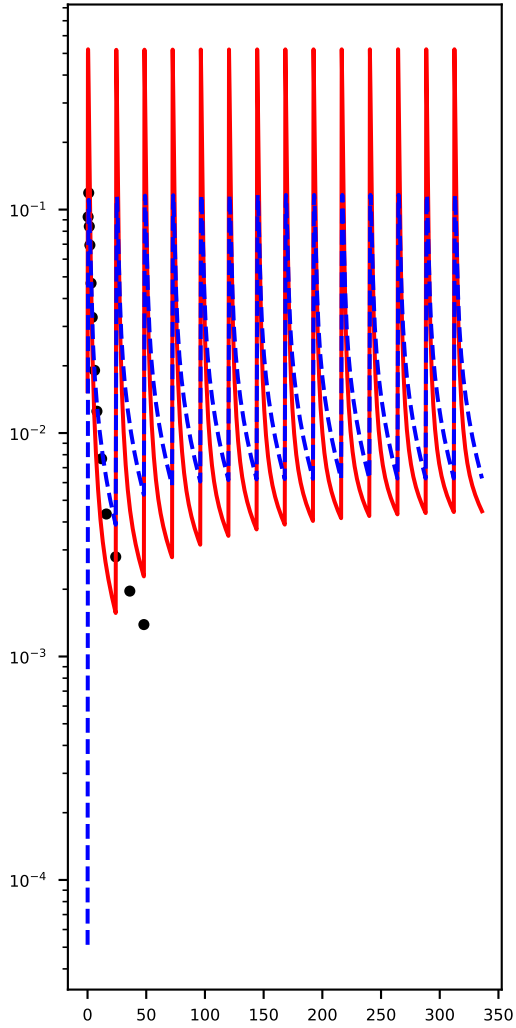
WHERE THE FITTED / FIXED PARAMETERS ENTER

- Permeability: the cellular permeability P in the permeability-limited exchange term.
- Lipophilicity (fixed): sets every tissue partition coefficient $K_{p,t}$ (the distribution terms in all organ ODEs).
- Fraction unbound (plasma, reference value) (fixed): f_u – scales every unbound-driven term ($K_{p,t}$, CL_{int} , GFR).
- GFR fraction (fixed): the renal filtration sink GFR in the kidney ODE.
- Specific intestinal permeability (transcellular) (fixed): P_{int} in the oral-absorption flux J_{abs} .

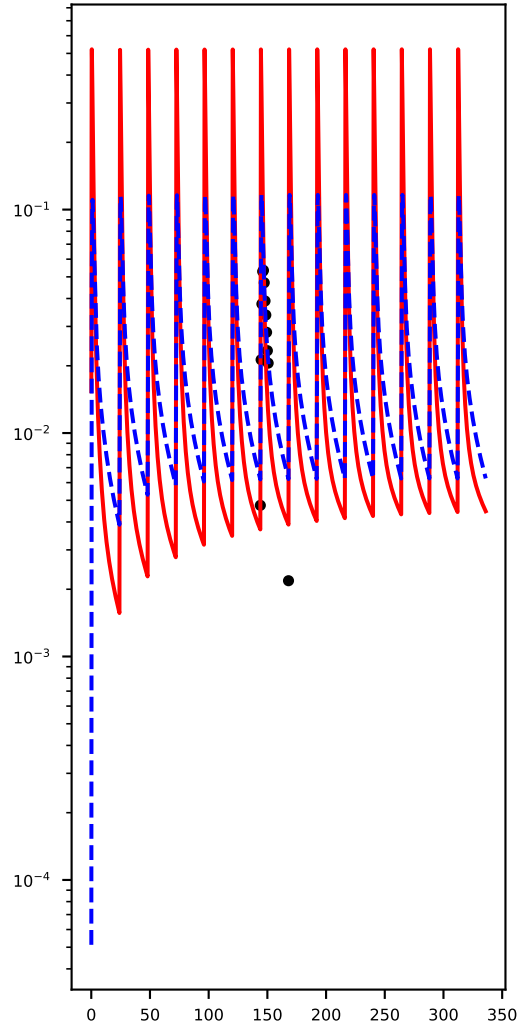
These are the governing equations; PK-Sim generates the fully instantiated system (every organ and sub-compartment, and the exact PK-Sim Standard K_p formula) at build time and integrates it numerically. The concentrations plotted are the peripheral-venous-plasma state.

Observed (●) vs agent (—) vs reference (--)

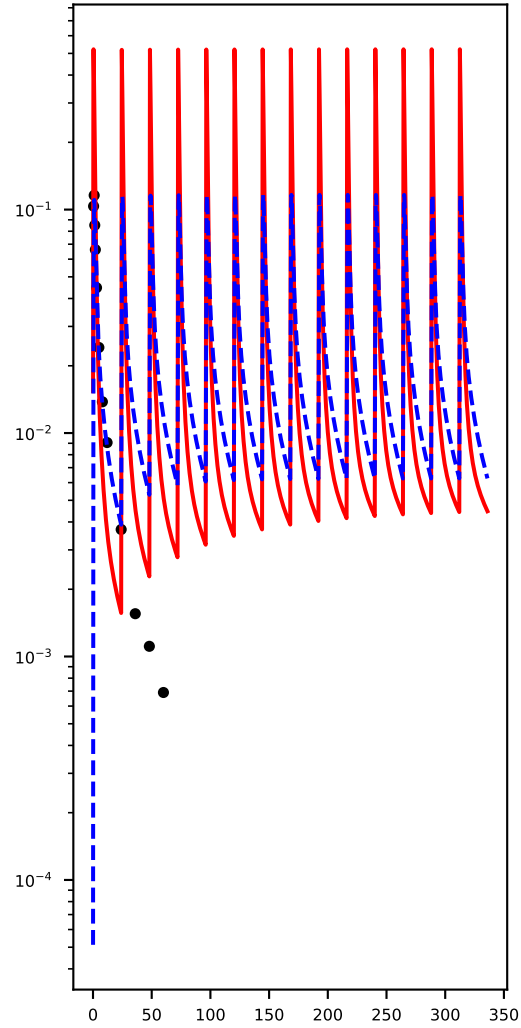
Kasichayanula 2011c - Healthy



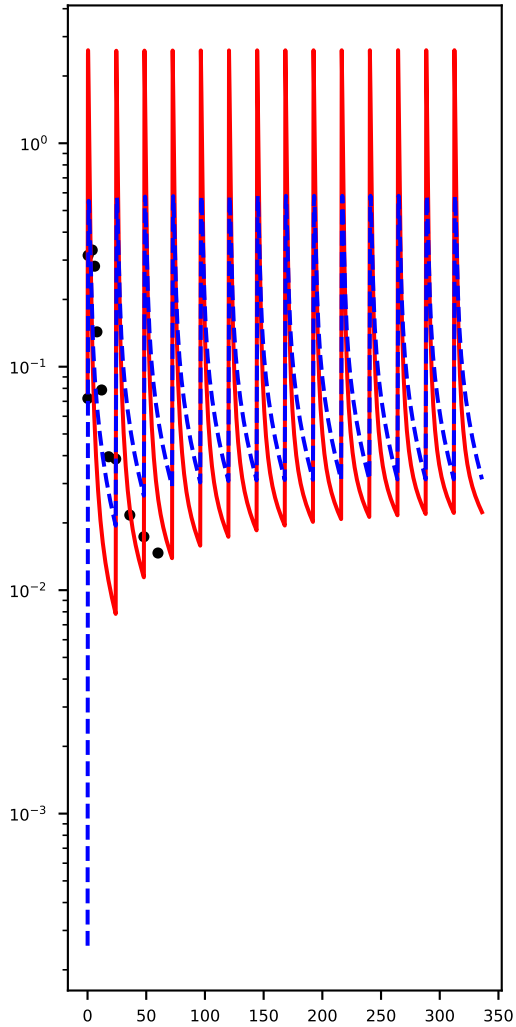
DeFronzo 2013 - Healthy Subjec



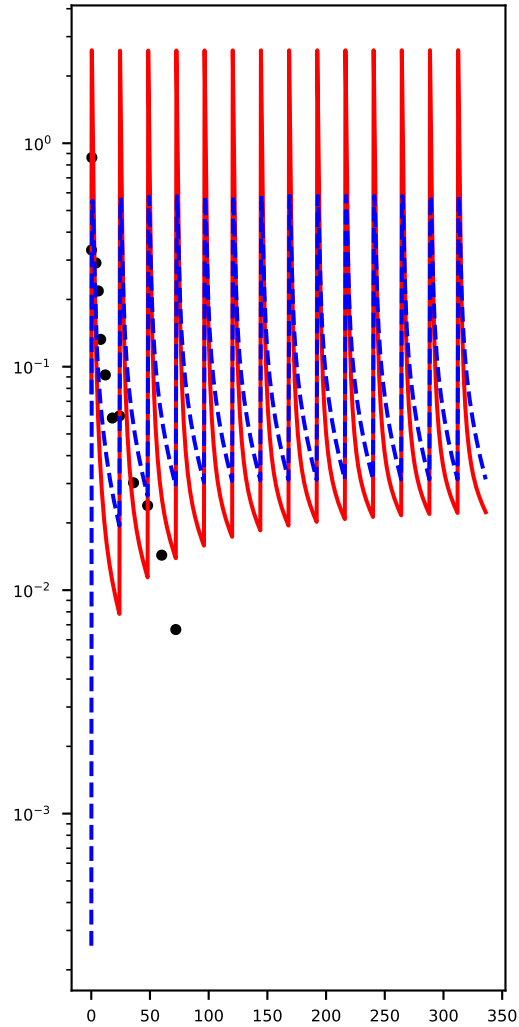
Vakkalagadda 2016 - Dapagliflo



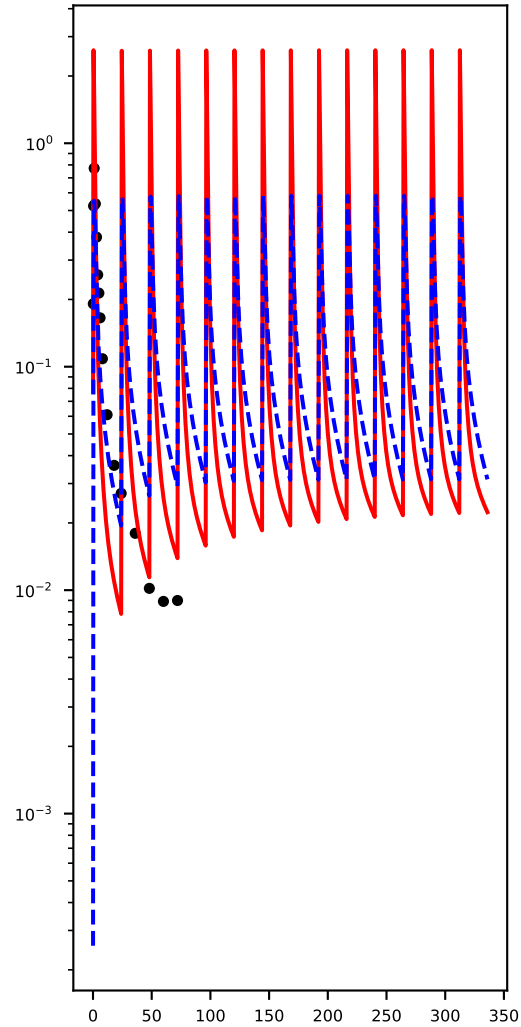
Kasichayanula 2013b - T2DM wit



Kasichayanula 2013b - T2DM wit

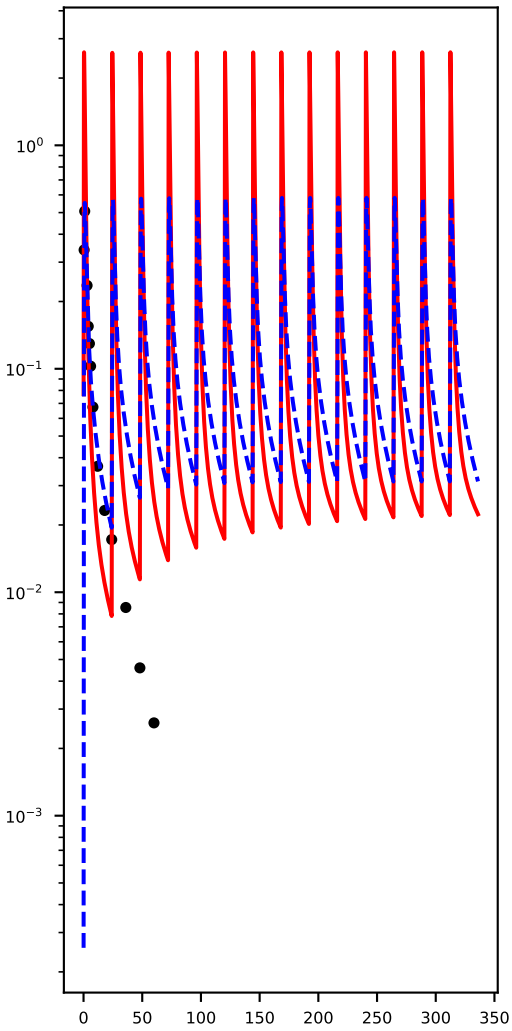


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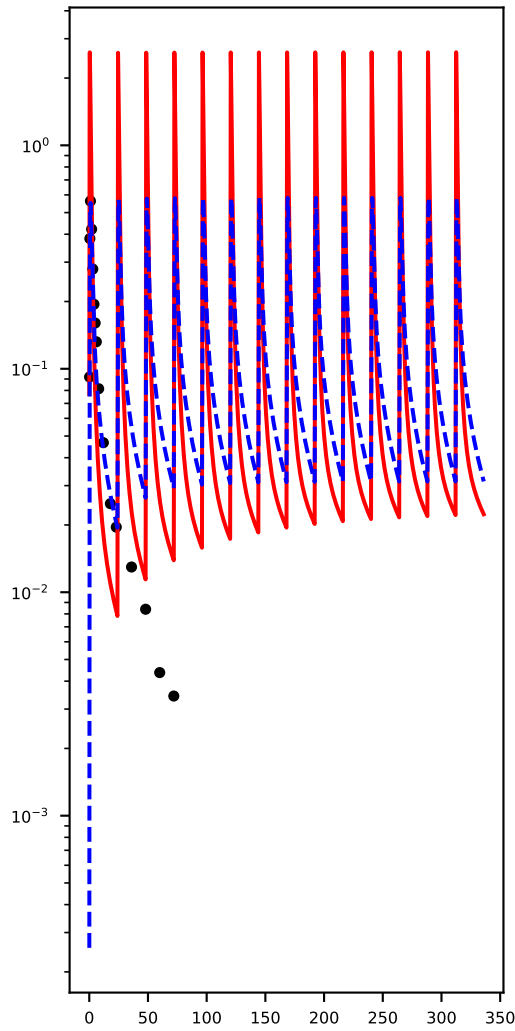


Observed (●) vs agent (—) vs reference (--)

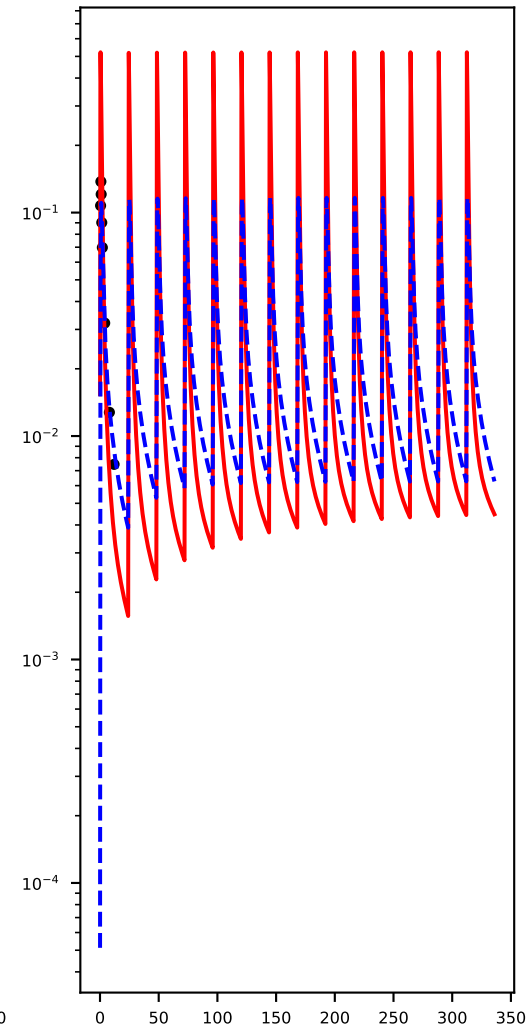
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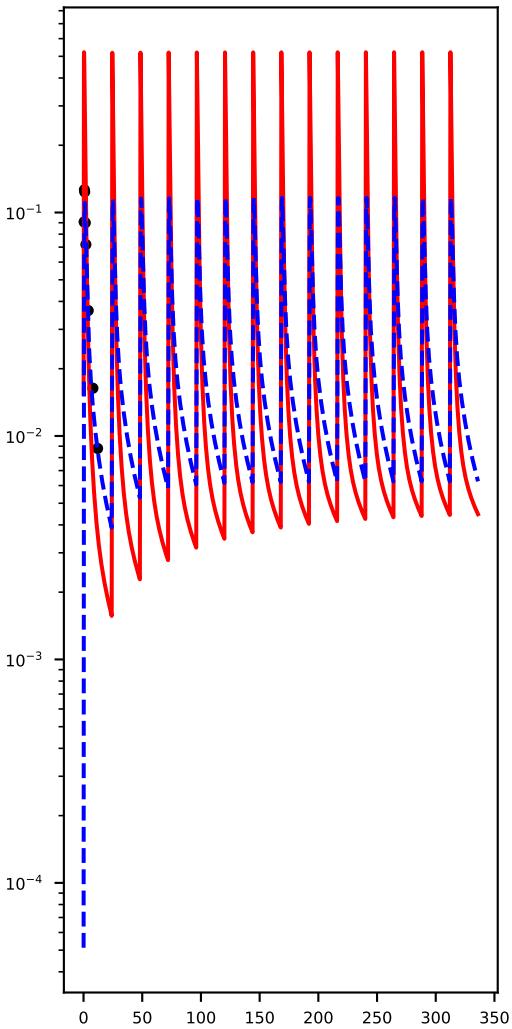
Kasichayanula 2013b - Healthy



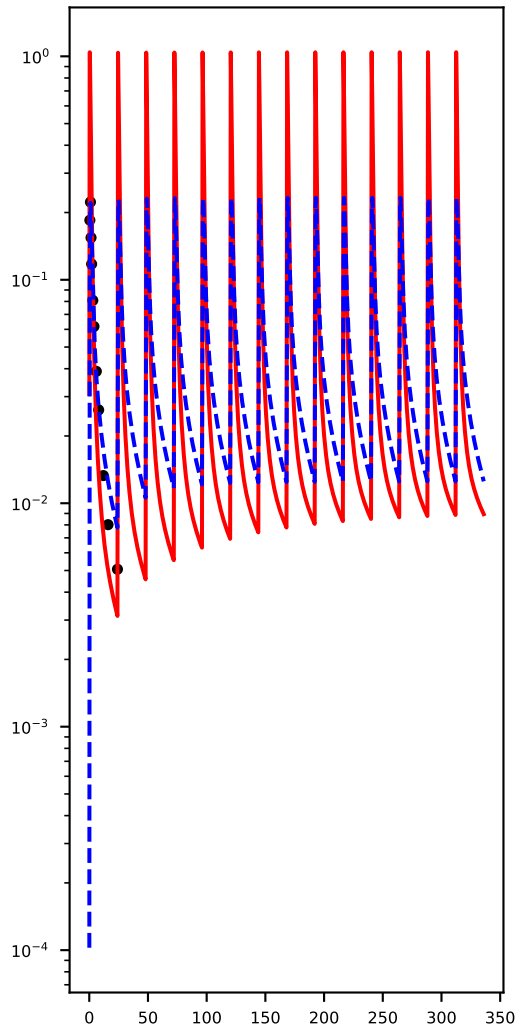
Kasichayanula 2013a - Study 2:



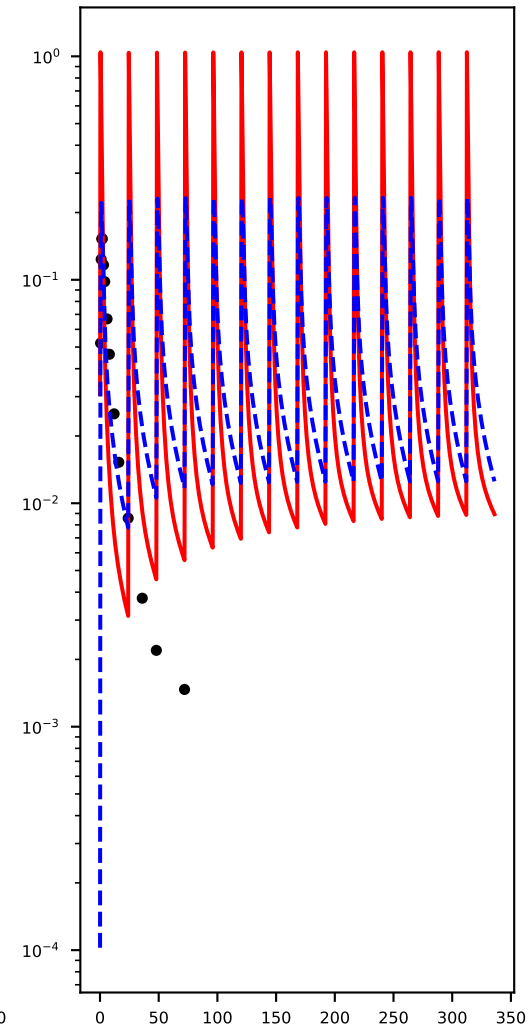
Kasichayanula 2013a - Study 1:



Kasichayanula 2012 - Study 1:

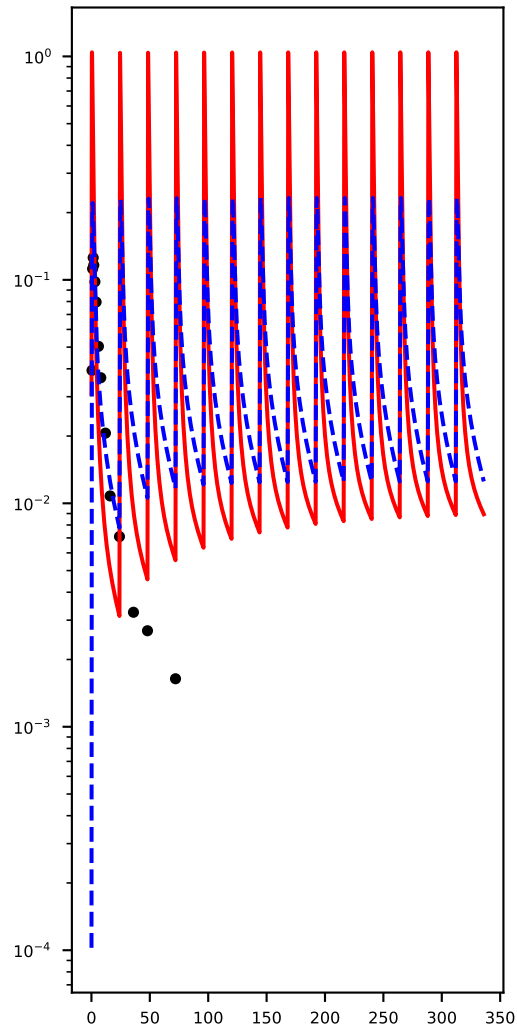


Kasichayanula 2011b - Study 3:

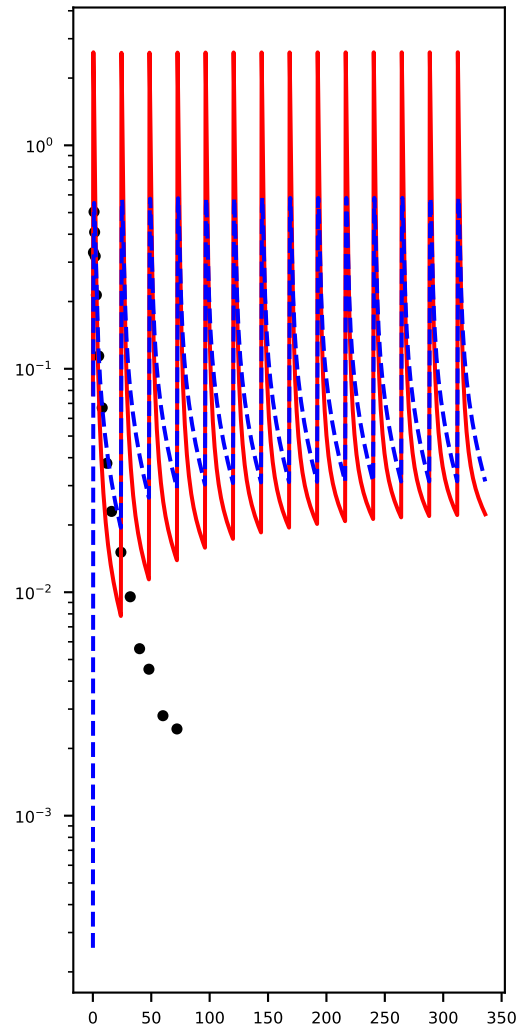


Observed (●) vs agent (—) vs reference (--)

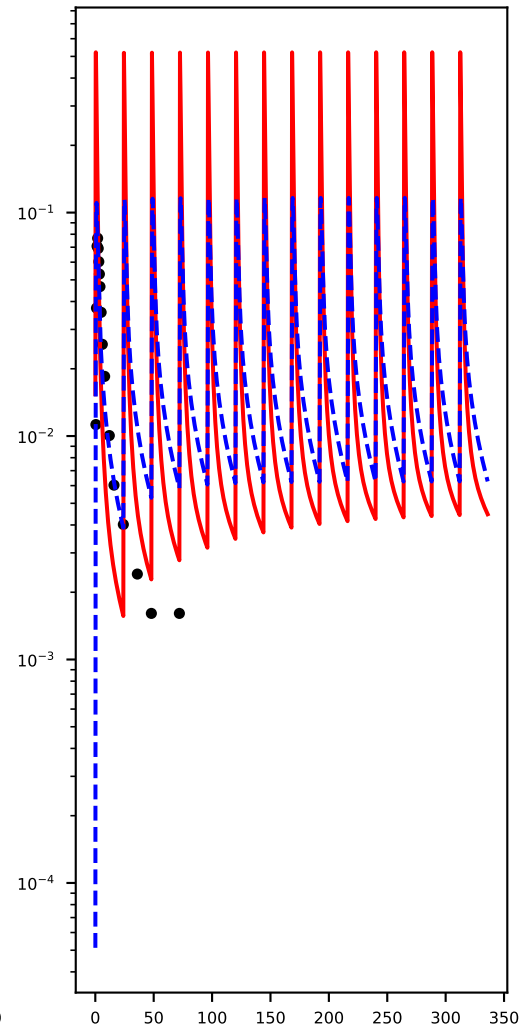
Kasichayanula 2011b - Study 2:



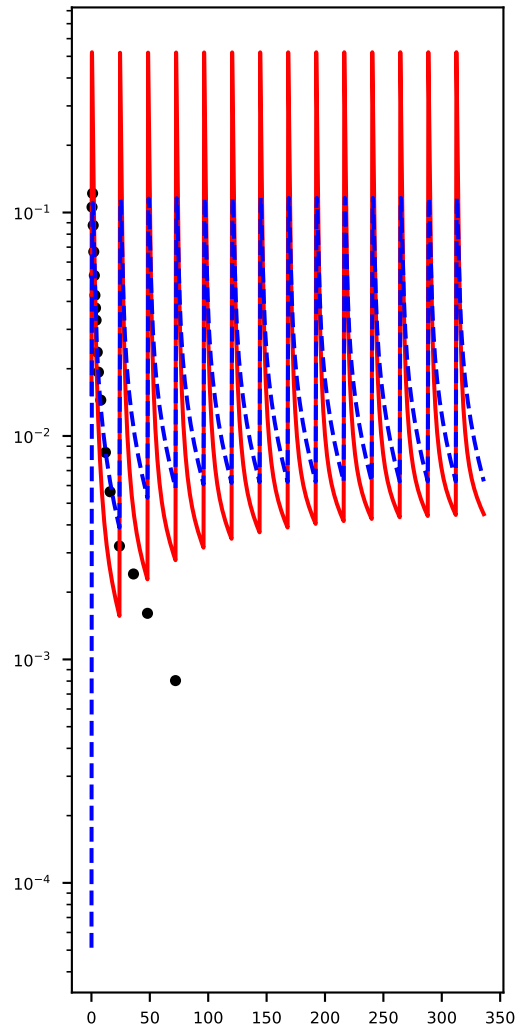
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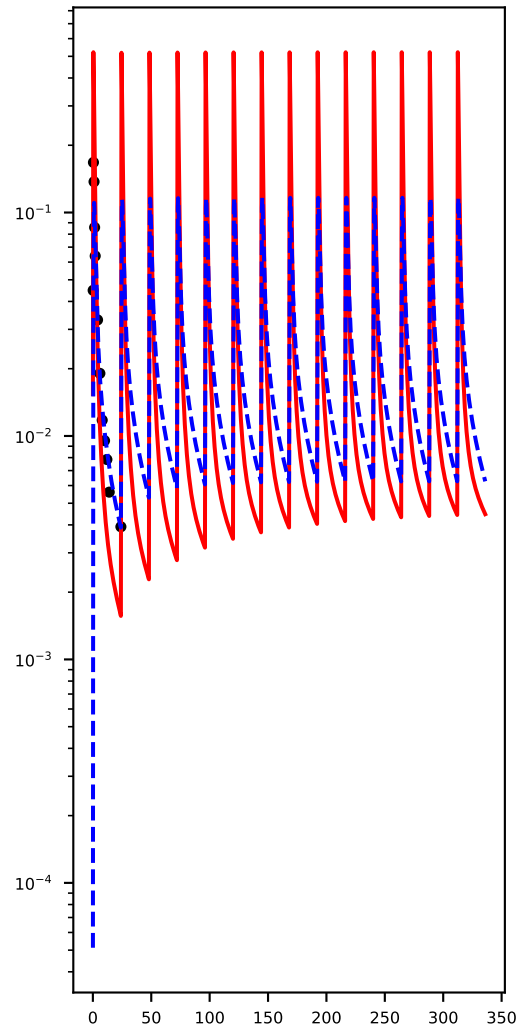
Kasichayanula 2011a - fed - Da



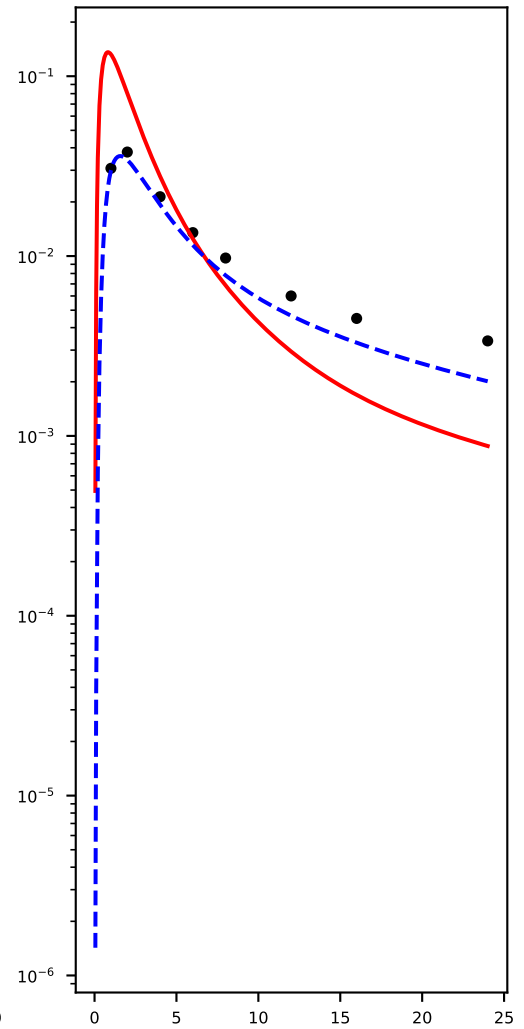
Kasichayanula 2011a - fasted -



Imamura 2013 - Control (Perpet)

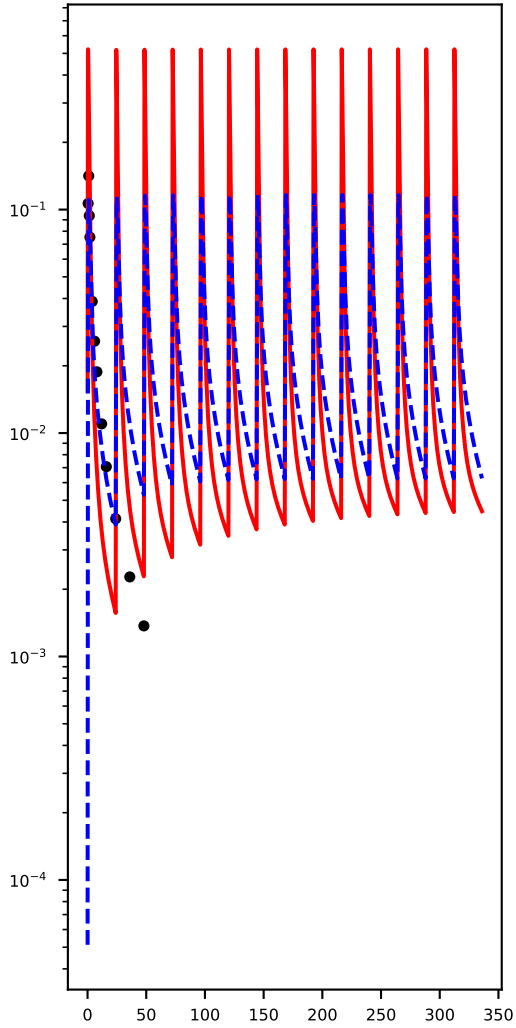


Chang 2015 - Study 1 Treatment

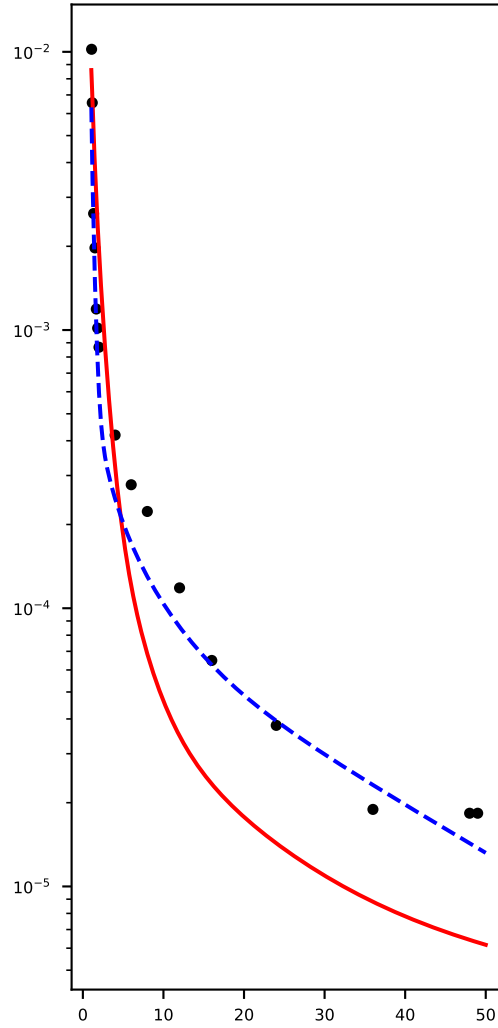


Observed (●) vs agent (—) vs reference (--)

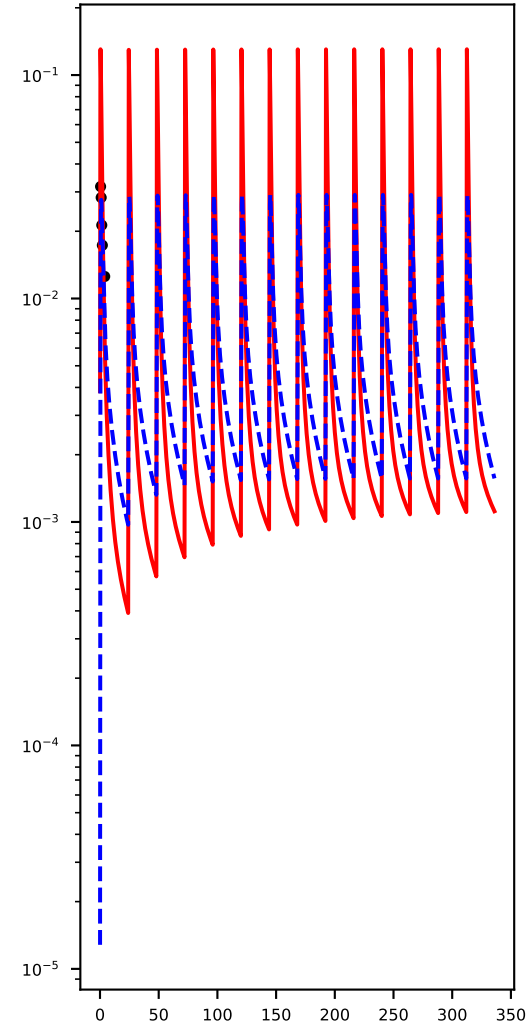
Boulton 2013 - Dapagliflozin p



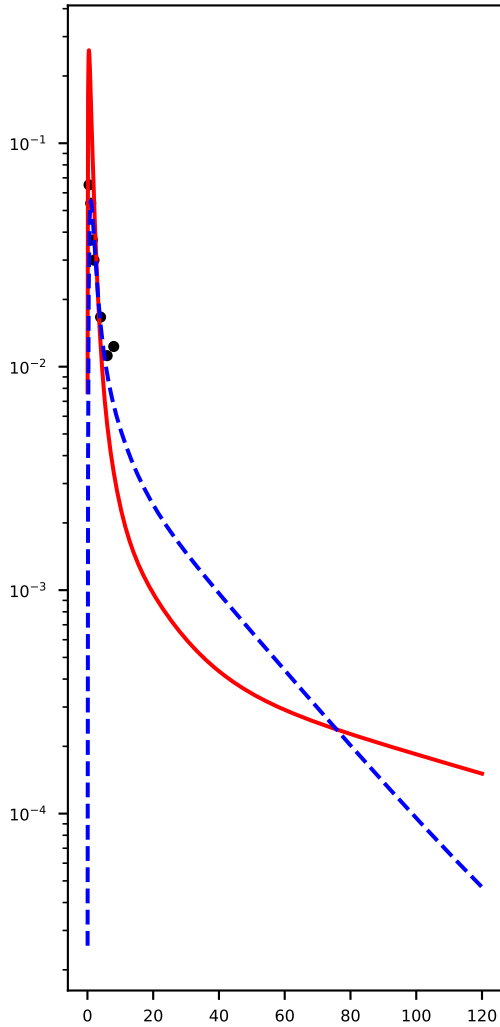
Boulton 2013 - 14C-dapaglifloz



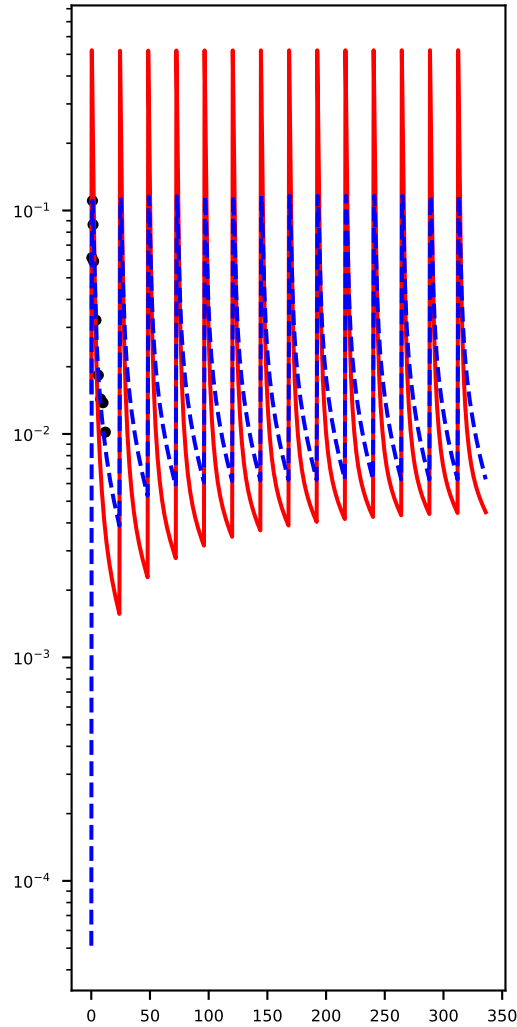
Komoroski 2009 - SAD 2.5 mg -



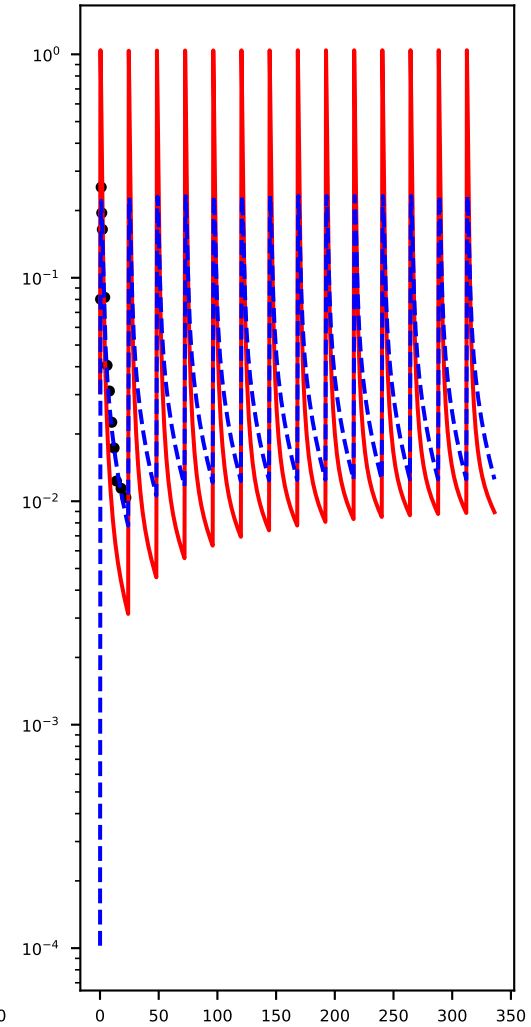
Komoroski 2009 - SAD 5 mg - Da



Komoroski 2009 - SAD 10 mg - D

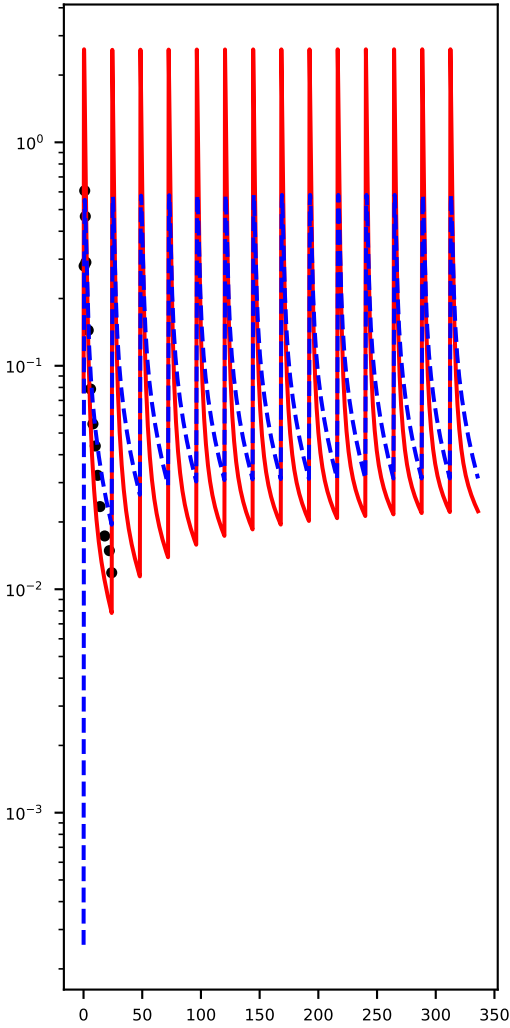


Komoroski 2009 - SAD 20 mg - D

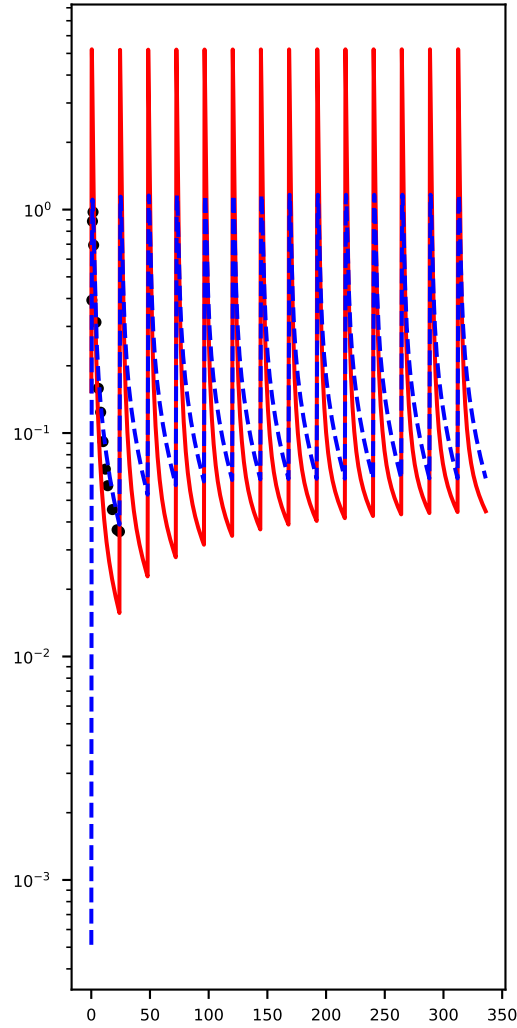


Observed (●) vs agent (—) vs reference (--)

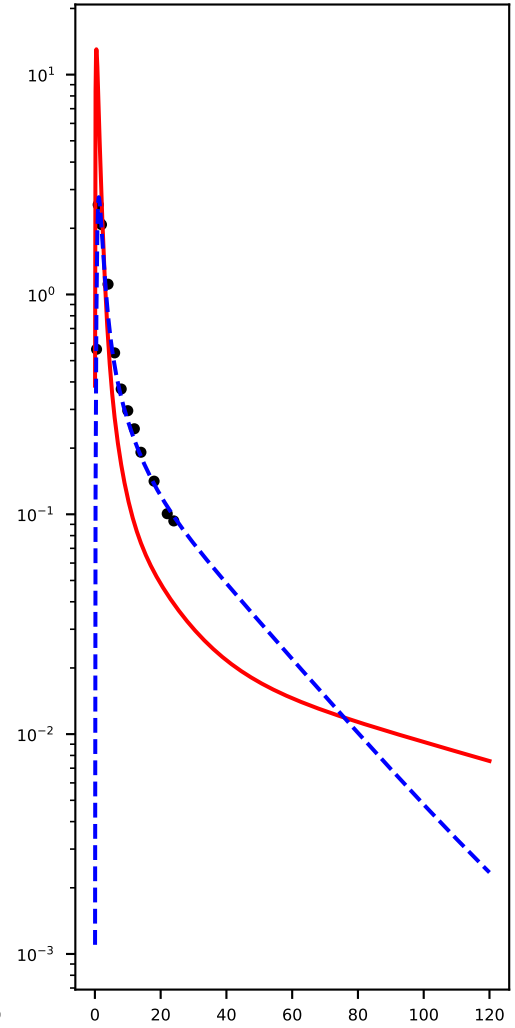
Komoroski 2009 - SAD 50 mg - D



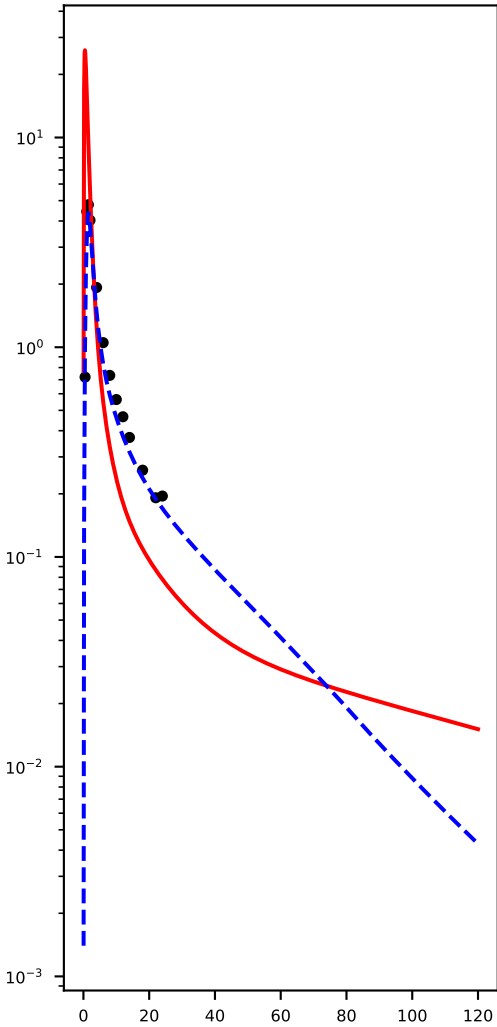
Komoroski 2009 - SAD 100 mg -



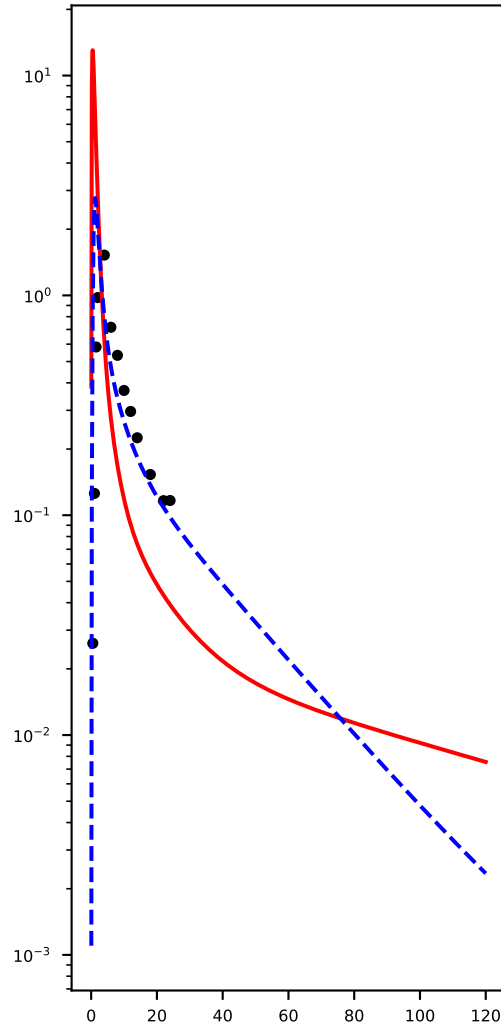
Komoroski 2009 - SAD 250 mg fa



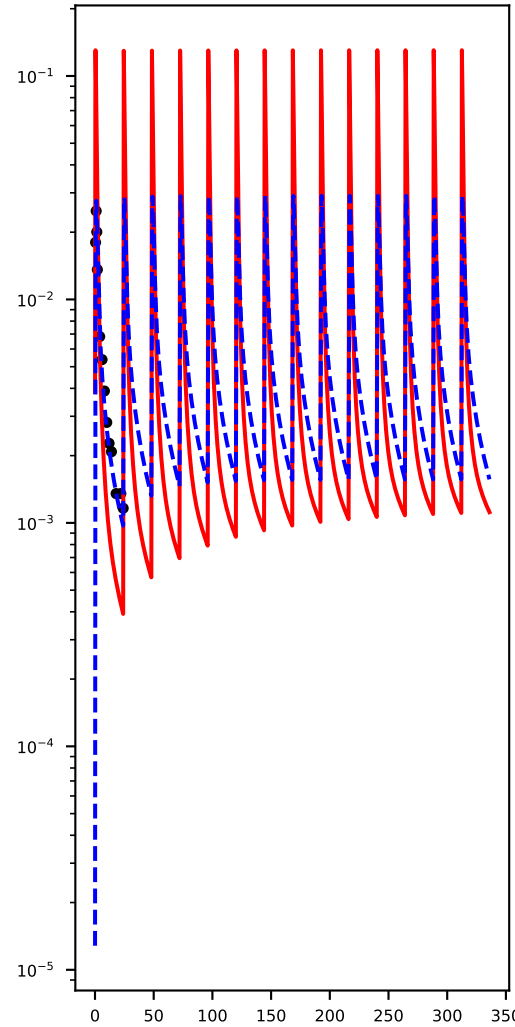
Komoroski 2009 - SAD 500 mg -



Komoroski 2009 - SAD 250 mg fe

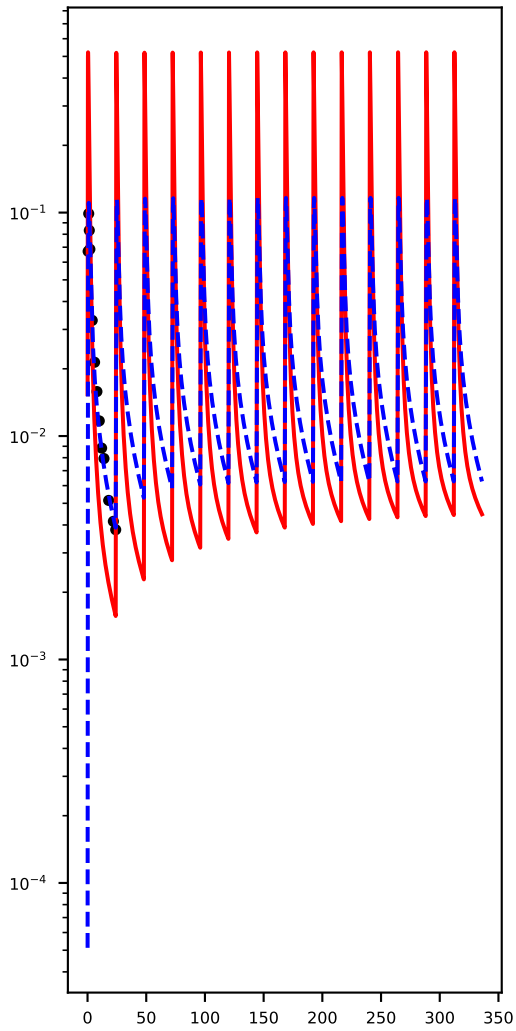


Komoroski 2009 - MAD 2.5 mg (d

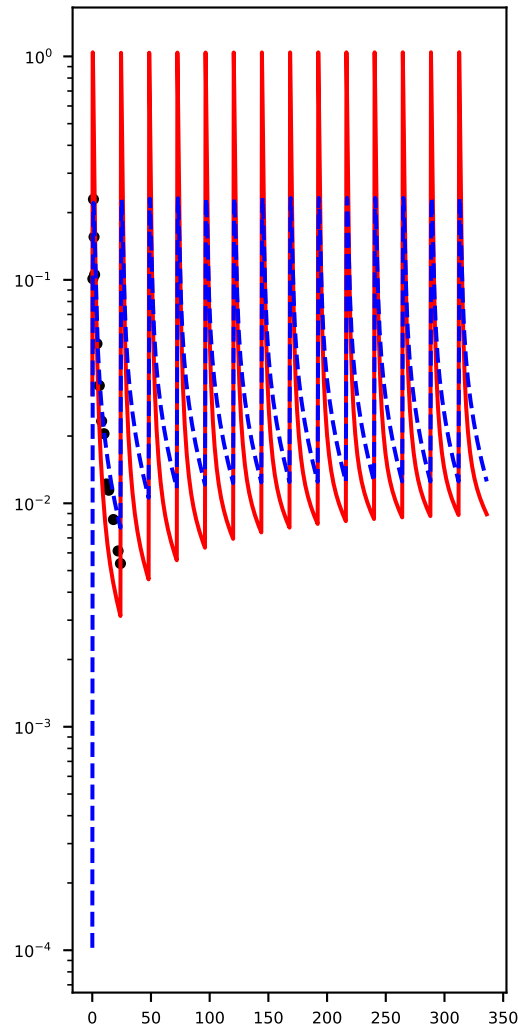


Observed (●) vs agent (—) vs reference (--)

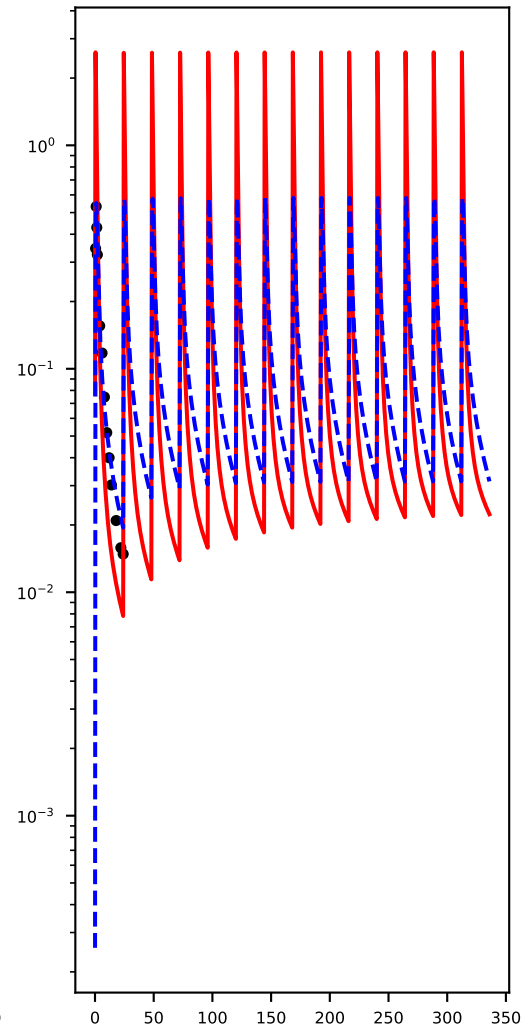
Komoroski 2009 - MAD 10 mg (da)



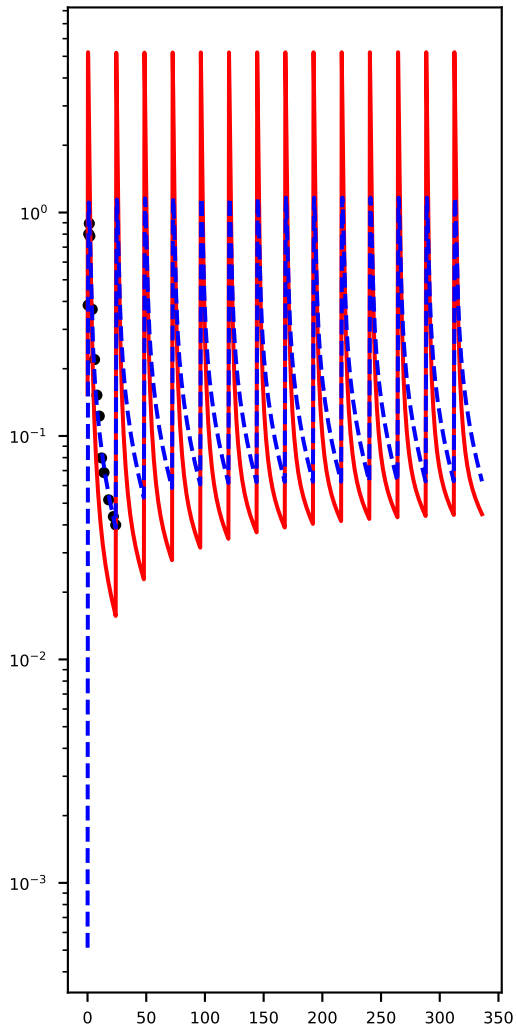
Komoroski 2009 - MAD 20 mg (da)



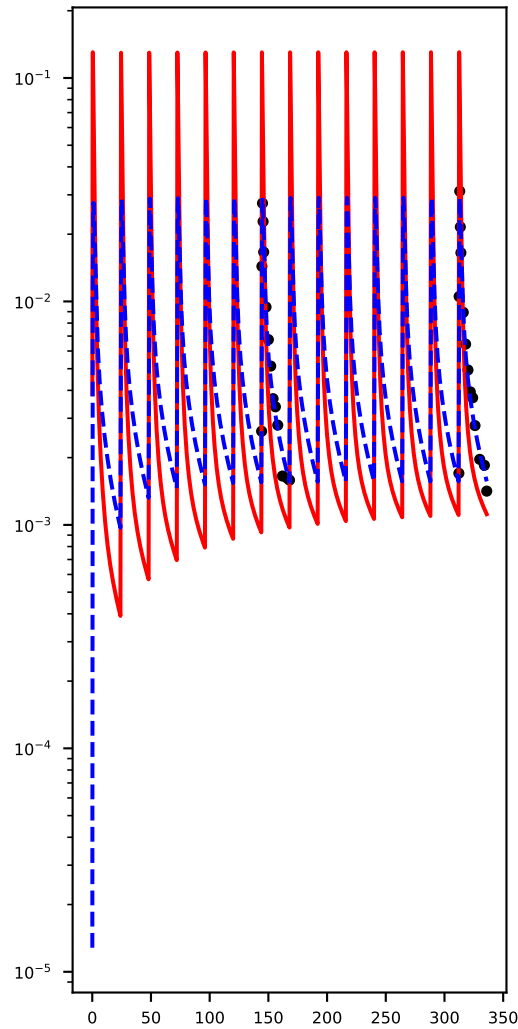
Komoroski 2009 - MAD 50 mg (da)



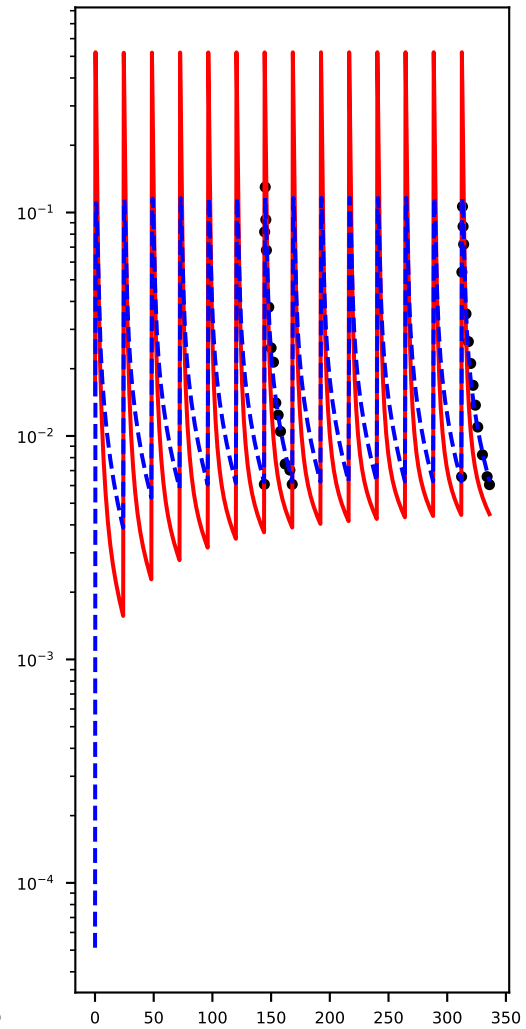
Komoroski 2009 - MAD 100 mg (d)



Komoroski 2009 - MAD 2.5 mg (d)

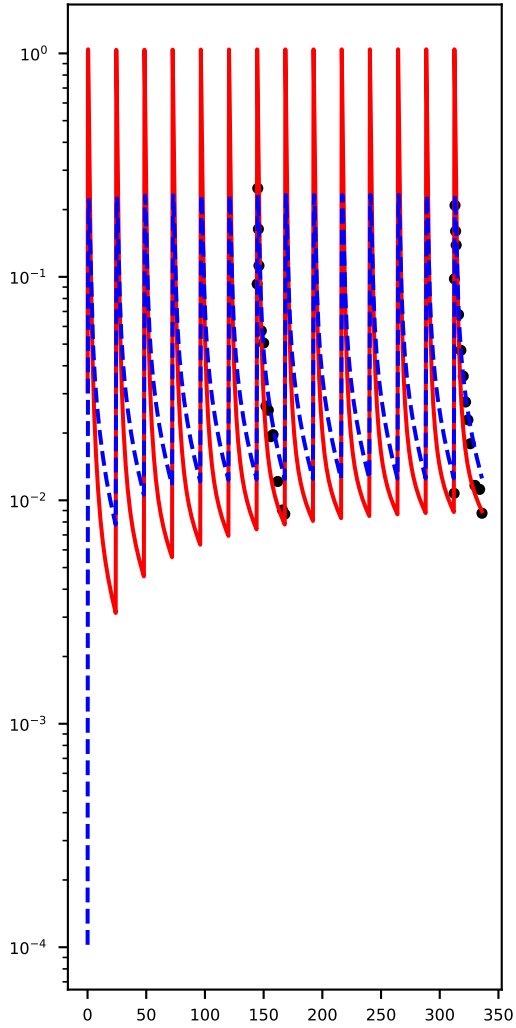


Komoroski 2009 - MAD 10 mg (da)

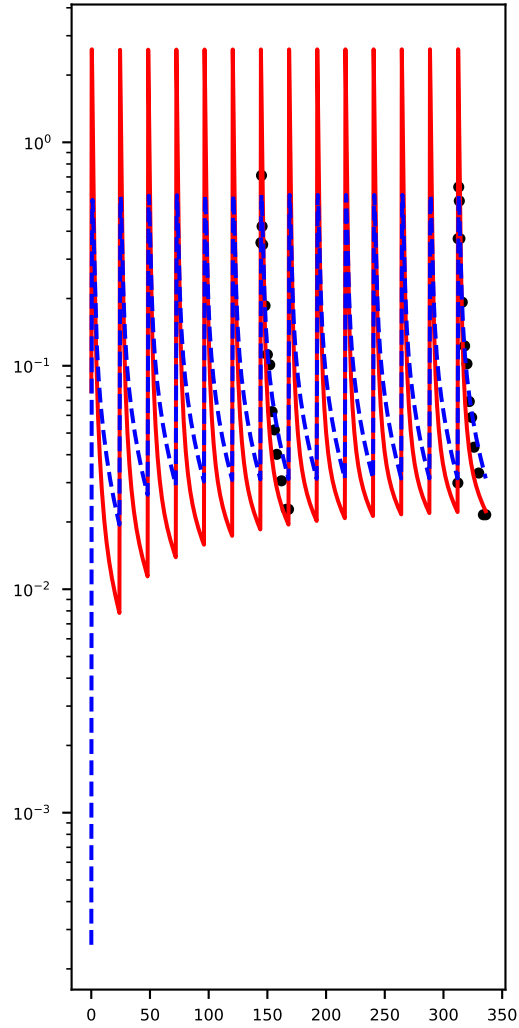


Observed (●) vs agent (—) vs reference (--)

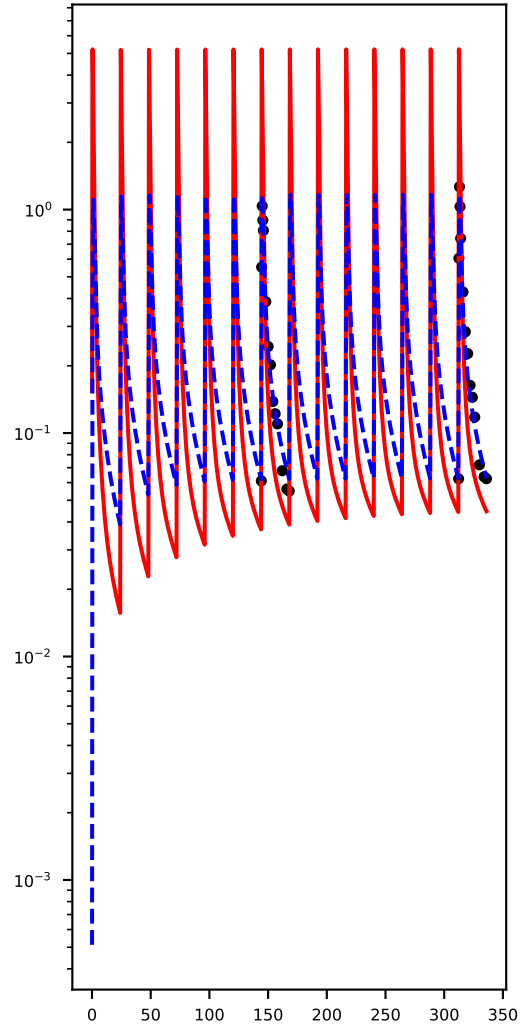
Komoroski 2009 - MAD 20 mg (da)



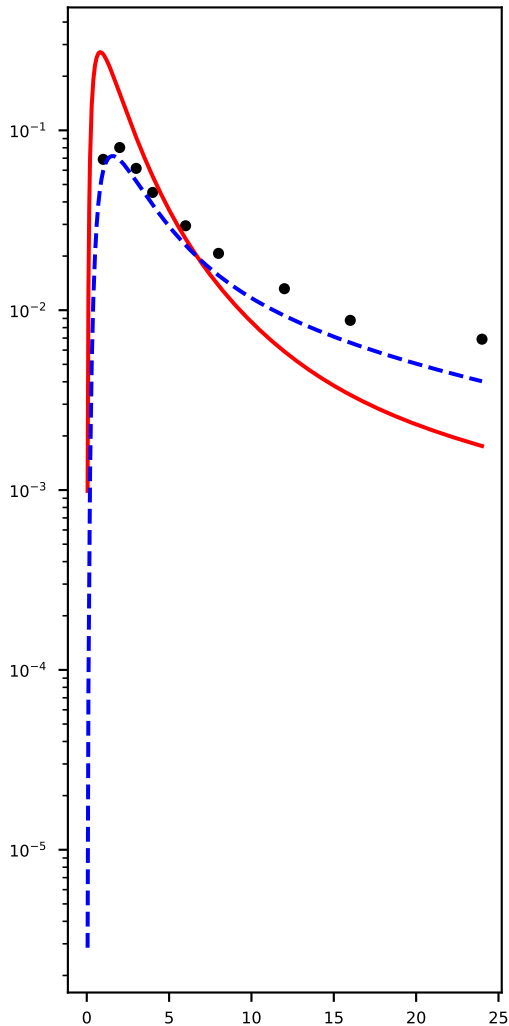
Komoroski 2009 - MAD 50 mg (da)



Komoroski 2009 - MAD 100 mg (d)



Chang 2015 - Study 2 Treatment



Ground-truth comparison & conclusion

Post-hoc evaluation vs the reference model (not seen during modeling). Each parameter is judged on distance from truth AND identifiability - a value far from truth but weakly identified is not a real error.

Reference GMFE 1.413 vs this model 2.468

SUMMARY

The structure DIFFERS from the ground-truth model in: partition method. Overall fit is comparable (this model GMFE 2.468 vs reference 1.413). Of 8 compared parameters, 3 recovered well, 5 differ but are weakly identified / minor, and 0 are off despite being influential. 2 of the fitted parameters are collinear (trade off against a partner), so the data constrain their combination but not their individual values - the split needs an in-vitro anchor rather than more plasma data.

STRUCTURE (this model | ground truth | match)

partition method: PK-Sim Standard | Rodgers and Rowland | DIFFERS
permeability method: PK-Sim Standard | PK-Sim Standard | match
processes: ['GlomerularFiltration', 'Hepatic-CYP', 'UGT1A9', 'UGT2B7'] | ['GlomerularFiltration', 'Hepatic-CYP', 'UGT1A9', 'UGT2B7'] | match

PARAMETERS (this | truth | fold | sensitivity | assessment)

CLspec/[Enzyme]@UGT1A9: 0.3946 | 0.3994 | 0.99x | 1 | [GOOD] recovered well (0.99x)
CLspec/[Enzyme]@UGT2B7: 0.01885 | 0.006604 | 2.9x | 0.081 | [MINOR] off 2.9x, but collinear with Permeability (trade-off 0.89): only their combination is identifiable from plasma parent data, so the split between them is not an independent fitting error - an in-vitro anchor (CLint, fraction metabolized) is needed to apportion it
Permeability: 2.89e-05 | 0.0003753 | 0.077x | 0.13 | [MINOR] off 0.077x, but collinear with CLspec/[Enzyme]@UGT2B7 (trade-off 0.89): only their combination is identifiable from plasma parent data, so the split between them is not an independent fitting error - an in-vitro anchor (CLint, fraction metabolized) is needed to apportion it
Lipophilicity: 2 | 2.672 | 0.75x | - | [GOOD] fixed, matches the reference value (0.75x)
Fraction unbound (plasma, reference value): 0.09 | None | - | - | no reference value
Solubility at reference pH: 100 | 0.221 | 4.5e+02x | - | [MINOR] fixed that differs from the reference (4.5e+02x); it was not estimated, so this is a prior choice rather than a fitting error
Reference pH: 7 | None | - | - | no reference value
Molecular weight: 408.9 | None | - | - | no reference value
GFR fraction: 1 | 0.79 | 1.3x | - | [GOOD] fixed, matches the reference value (1.3x)
CLspec/[Enzyme]@Hepatic-CYP: 0.005 | 0.1433 | 0.035x | - | [MINOR] fixed that differs from the reference (0.035x); it was not estimated, so this is a prior choice rather than a fitting error
Specific intestinal permeability (transcellular): 0.001 | 3.968e-05 | 25x | - | [MINOR] fixed that differs from the reference (25x); it was not estimated, so this is a prior choice rather than a fitting error
Cl: 1 | None | - | - | no reference value

CONCLUSION

The model does NOT yet adequately reproduce the observed plasma concentrations: overall GMFE is 2.468 (beyond the ~2-fold typical error). This indicates a structural or parameter problem to

Ground-truth comparison & conclusion

resolve (check per-route bias and any parameters pinned to a bound) before the model can be considered fit for purpose.

Full modeling trajectory (LLM)

[Step 1] osp_inspect, osp_options

-> task description: objective, known biology, literature priors, current model, and the observed clinical data | authoritative action space: what you may edit and legal choices

[Step 2] osp_optimize

-> optimized ['Specific intestinal permeability (transcellular)', 'Permeability', 'CLspec/[Enzyme]@UGT1A9', 'CLspec/[Enzyme]@UGT2B7', 'CLspec/[Enzyme]@Hepatic-CYP'] on 4 study(ies) -> GMFE 3.107 (best so far 3.107) | NEXT: FIX one of ('Specific intestinal permeability (transcellular)', 'Permeability') - prefer the one with a literature anchor - and estimate only the other, so the fit is well-conditioned instead of trading the two off (+2 more identifiability note(s)) optimized={'Specific intestinal permeability (transcellular)': 4.3473345660757765e-06, 'Permeability': 0.00021966557503841667, 'CLspec/[Enzyme]@UGT1A9': 0.03885346927328239, 'CLspec/[Enzyme]@UGT2B7': 0.012892047982017303, 'CLspec/[Enzyme]@Hepatic-CYP': 0.01927270721513828}

[Step 3] osp_optimize

-> optimized ['Specific intestinal permeability (transcellular)', 'CLspec/[Enzyme]@UGT1A9'] on 4 study(ies) -> GMFE 3.1 (best so far 3.1) optimized={'Specific intestinal permeability (transcellular)': 4.987362893589893e-06, 'CLspec/[Enzyme]@UGT1A9': 0.04306213593276214}

[Step 4] osp_optimize

-> optimized ['CLspec/[Enzyme]@UGT1A9', 'CLspec/[Enzyme]@UGT2B7', 'Specific intestinal permeability (transcellular)', 'Permeability'] on 4 study(ies) -> GMFE 3.258 (best so far 3.1) | NEXT: FIX 'Specific intestinal permeability (transcellular)' to a known/literature value (use the given literature anchor, or your own pharmacological knowledge, e.g. a measured logP/logD for lipophilicity, a reported pKa) instead of estimating it, then refit only the parameters the data actually constrain (+1 more identifiability note(s)) optimized={'CLspec/[Enzyme]@UGT1A9': 0.6935277296846187, 'CLspec/[Enzyme]@UGT2B7': 0.049281444781828394, 'Specific intestinal permeability (transcellular)': 0.001, 'Permeability': 2.5158535281611447e-05} at_bound=[{'parameter': 'Specific intestinal permeability (transcellular)', 'value': 0.001, 'bound': 'upper'}]

[Step 5] osp_optimize

-> optimized ['CLspec/[Enzyme]@UGT1A9', 'CLspec/[Enzyme]@UGT2B7', 'Permeability'] on 4 study(ies) -> GMFE 2.468 (best so far 2.468) | NEXT: FIX 'CLspec/[Enzyme]@UGT2B7' to a known/literature value (use the given literature anchor, or your own pharmacological knowledge, e.g. a measured logP/logD for lipophilicity, a reported pKa) instead of estimating it, then refit only the parameters the data actually constrain optimized={'CLspec/[Enzyme]@UGT1A9': 0.3946448242915066, 'CLspec/[Enzyme]@UGT2B7': 0.018850874596835872, 'Permeability': 2.8904303298160844e-05}

[Step 6] osp_optimize

-> optimized ['CLspec/[Enzyme]@UGT1A9', 'Permeability'] on 4 study(ies) -> GMFE 2.615 (best so far 2.468) optimized={'CLspec/[Enzyme]@UGT1A9': 0.16018517068598345, 'Permeability': 2.2752611430815068e-05}